

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 4

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Score:	_____	Date:	_____	Instructor's Signature:	_____

Lab # 4: To Configure Firewall Hot Standby

Objective: The objective of this lab To configure Firewall Hot Standby.

Duration: 3 Hours

Expected Outcomes: By the end of this lab session, students will be able to:

1. Understand the basic principles of hot standby
2. Understand the VGMP and HRP Protocols
3. Master the configuration of Hot Standby using the CLI

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions.

Scenario

An enterprise needs to provide uninterrupted services. To avoid line interruption caused by network devices or other external factors, the enterprise wants to implement redundancy at the network egress to increase network reliability.

In this lab, two firewalls are deployed as gateways at the network egress to ensure smooth communication between the internal network and the external network in the case of a single-node fault. **Topology**

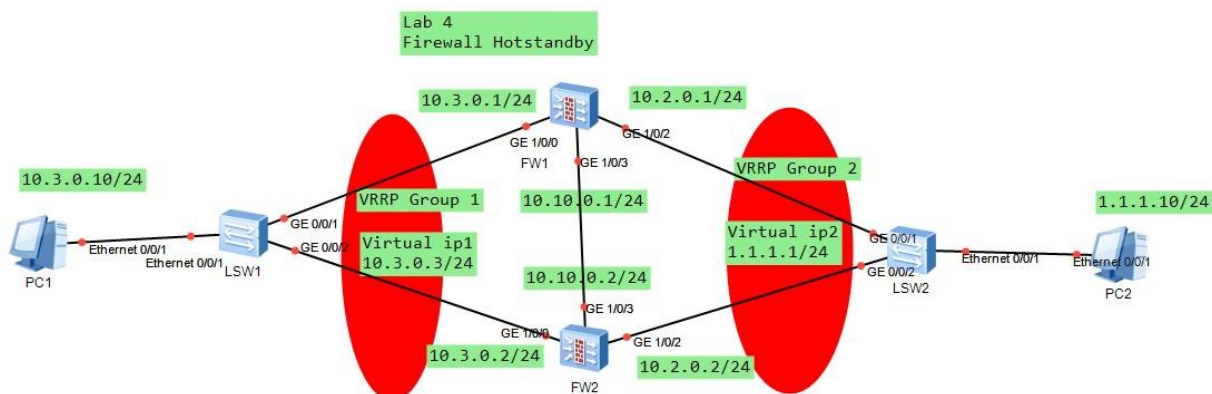


Figure 1 Firewall Hot Standby configuration

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HRP (Huawei Redundancy Protocol)

HRP is a proprietary protocol used by Huawei firewalls to manage the state synchronization between two devices in a high-availability (HA) cluster. Its primary role is to ensure that the standby device is always ready to take over the active role without any loss of active connections.

Key Functions of HRP:

- **State Synchronization:** HRP is responsible for synchronizing a wide range of state information from the active firewall to the standby firewall. This includes:
 - **Session Table:** Active connections (TCP, UDP, etc.) are mirrored to the standby device. This is critical for preventing session loss during a failover. For example, if a user is downloading a file when the active firewall fails, the session information on the standby device allows the download to continue without interruption.
 - **NAT Table:** Network Address Translation (NAT) session information is synchronized.
 - **Security Policy and Configuration:** HRP ensures that the security policies, NAT policies, and other configurations are consistent across both devices. In an active/standby setup, configurations made on the active device are automatically synchronized to the standby.
- **Heartbeat Mechanism:** HRP uses a dedicated **heartbeat interface** (or multiple interfaces) to monitor the health of the peer device. If the active device fails to send heartbeat packets within a configured time, the standby device assumes the active role.
- **Active/Standby Role Management:** While HRP itself doesn't make the failover decision, it works in conjunction with VRRP to manage the roles of the firewalls. The HRP state of a firewall can be active, standby, or normal.

Both **VGMP (VRRP Group Management Protocol)** and **HRP (Huawei Redundancy Protocol)** are crucial for implementing high-availability and redundancy on Huawei firewalls like the USG 6000V. They work together to ensure that a backup device can take over seamlessly if the primary device fails.

VGMP (VRRP Group Management Protocol)

VGMP is a protocol that extends **VRRP (Virtual Router Redundancy Protocol)** to manage the state of multiple VRRP groups simultaneously. In a typical HA setup with a Huawei firewall, you'll have multiple VRRP groups, each associated with a different interface (e.g., one for the

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upstream link, one for the downstream link). The VGMP ensures that all VRRP groups on a single firewall operate in a coordinated manner.

Key Functions of VGMP:

- **VRRP Group Coordination:** VGMP groups multiple VRRP instances and manages them as a single entity. It prevents a scenario where one VRRP group on a firewall is active while another is standby, which could cause routing issues. Instead, it ensures that a single firewall is either the active router for all its VRRP groups or the standby for all of them.
- **Active/Standby Role Transition:** VGMP is the protocol that triggers the failover. When the active firewall experiences a fault (e.g., an interface goes down, the heartbeat fails), VGMP detects this and triggers a state change. It forces the local VRRP groups to transition from the active state to the standby state and signals the peer firewall to become active.
- **Health Monitoring:** VGMP monitors the health of tracked interfaces and other system parameters. If a tracked object fails, VGMP will change the firewall's state from active to standby, initiating a failover to the healthy device.

Lab Planning

Device	Interface	IP Address	Security Zone
FW1	GigabitEthernet 1/0/0	10.3.0.1/24	Trust
	GigabitEthernet 1/0/2	10.2.0.1/24	Untrust
	GigabitEthernet 1/0/3	10.10.0.1/24	DMZ
FW2	GigabitEthernet 1/0/0	10.3.0.2/24	Trust
	GigabitEthernet 1/0/2	10.2.0.2/24	Untrust
	GigabitEthernet 1/0/3	10.10.0.2/24	DMZ
PC1	Eth 0/0/1	10.3.0.10/24	Trust
PC2	Eth 0/0/1	1.1.1.10/24	Untrust
SW1	GigabitEthernet 0/0/1 GigabitEthernet 0/0/2 Ethernet 0/0/1	Acess	PVID: VLAN 10
SW2	GigabitEthernet 0/0/1 GigabitEthernet 0/0/2 Ethernet 0/0/1	Acess	PVID: VLAN 10

Configuration Roadmap

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1. Configure the basic IP addresses and security zones on FW1 and FW2, and apply the relevant security policies.
2. Configure the hot standby. FW1 functions as the active node and FW2 functions as the standby node.

Configuration Procedure on the CLI

Step 1 Configure the upstream and downstream service interfaces of FW1 and FW2. Configure the IP addresses for interfaces and add them to security zones.

Configure IP addresses for the upstream and downstream service interfaces of FW1.

```
<FW1> system-view
[FW1] interface GigabitEthernet1/0/0
[FW1-GigabitEthernet1/0/0] ip address 10.3.0.1 255.255.255.0
[FW1-GigabitEthernet1/0/0] quit
[FW1] interface GigabitEthernet1/0/2
[FW1-GigabitEthernet1/0/2] ip address 10.2.0.1 255.255.255.0
[FW1-GigabitEthernet1/0/2] quit
```

Configure VRRP group 1 on GigabitEthernet1/0/0 of FW1 and add it to the VGMP group in the Active state.

```
[FW1] interface GigabitEthernet1/0/0
[FW1-GigabitEthernet1/0/0] vrrp vrid 1 virtual-ip 10.3.0.3 255.255.255.0 active [FW1-
GigabitEthernet1/0/0] quit
```

Configure VRRP group 2 on GigabitEthernet1/0/2 of FW1 and add it to the VGMP group in the Active state.

```
[FW1] interface GigabitEthernet1/0/2
[FW1-GigabitEthernet1/0/2] vrrp vrid 2 virtual-ip 1.1.1.1 255.255.255.0 active [FW1-
GigabitEthernet1/0/2] quit
```

Add the upstream and downstream service interfaces of FW1 to security zones.

```
[FW1] firewall zone trust
[FW1-zone-trust] add interface GigabitEthernet1/0/0
[FW1-zone-trust] quit
[FW1] firewall zone untrust
[FW1-zone-untrust] add interface GigabitEthernet1/0/2
```

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[FW1-zone-untrust] quit

Configure the upstream and downstream service interfaces of FW2.

```
<FW2> system-view
[FW2] interface GigabitEthernet1/0/0
[FW2-GigabitEthernet1/0/0] ip address 10.3.0.2 255.255.255.0
[FW2-GigabitEthernet1/0/0] quit
[FW2] interface GigabitEthernet1/0/2
[FW2-GigabitEthernet1/0/2] ip address 10.2.0.2 255.255.255.0
[FW2-GigabitEthernet1/0/2] quit
```

Configure VRRP group 1 on GigabitEthernet1/0/0 of FW2 and add it to the VGMP group in the Standby state.

```
[FW2] interface GigabitEthernet1/0/0
[FW2-GigabitEthernet1/0/0] vrrp vrid 1 virtual-ip 10.3.0.3 255.255.255.0 standby [FW2-
GigabitEthernet1/0/0] quit
```

Configure VRRP group 2 on GigabitEthernet1/0/2 of FW2 and add it to the VGMP group in the Standby state.

```
[FW2] interface GigabitEthernet1/0/2
[FW2-GigabitEthernet1/0/2] vrrp vrid 2 virtual-ip 1.1.1.1 255.255.255.0 standby [FW2-
GigabitEthernet1/0/2] quit
```

Add the upstream and downstream service interfaces of FW2 to security zones.

```
[FW2] Firewall Zone Trust
[FW2-zone-trust] add interface GigabitEthernet1/0/0
[FW2-zone-trust] quit
[FW2] firewall zone untrust
[FW2-zone-untrust] add interface GigabitEthernet1/0/2
[FW2-zone-untrust] quit
```

Step 2 Configure the heartbeat cables for FW1 and FW2.

Configure an IP address for the heartbeat interface GigabitEthernet1/0/3 of FW1.

```
[FW1] interface GigabitEthernet1/0/3
```

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```
[FW1-GigabitEthernet1/0/3] ip address 10.10.0.1 255.255.255.0  
[FW1-GigabitEthernet1/0/3] quit
```

Configure an IP address for the heartbeat interface GigabitEthernet1/0/3 of FW2.

```
[FW2] interface GigabitEthernet1/0/3  
[FW2-GigabitEthernet1/0/3] ip address 10.10.0.2 255.255.255.0  
[FW2-GigabitEthernet1/0/3] quit
```

Add the heartbeat interface GigabitEthernet1/0/3 of FW1 to the DMZ.

```
[FW1] firewall zone dmz  
[FW1-zone-dmz] add interface GigabitEthernet1/0/3  
[FW1-zone-dmz] quit
```

Add the heartbeat interface GigabitEthernet1/0/3 of FW2 to the DMZ.

```
[FW2] firewall zone dmz  
[FW2-zone-dmz] add interface GigabitEthernet1/0/3  
[FW2-zone-dmz] quit
```

Configure the authentication key for the heartbeat interface of FW1 and enable the hot standby function.

```
[FW1] hrp interface GigabitEthernet1/0/3 remote 10.10.0.2  
[FW1] hrp enable
```

Configure the authentication key for the heartbeat interface of FW2 and enable the hot standby function.

```
[FW2] hrp interface GigabitEthernet1/0/3 remote 10.10.0.1  
[FW2] hrp enable
```

Step 3 Configure a security policy on FW1 to allow service packets to pass through. After hot standby is enabled, the security policy configured on FW1 will be automatically synchronized to FW2.

Configure a forwarding policy between the Trust and Untrust zones on FW1.

```
HRP_M[FW1] security-policy  
HRP_M[FW1-policy-security] rule name trust_to_untrust
```

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```
HRP_M[FW1-policy-security-rule-trust_to_untrust] source-zone trust
HRP_M[FW1-policy-security-rule-trust_to_untrust] destination-zone untrust
HRP_M[FW1-policy-security-rule-trust_to_untrust] source-address 10.3.0.0 24
HRP_M[FW1-policy-security-rule-trust_to_untrust] action permit
HRP_M[FW1-policy-security-rule-trust_to_untrust] quit
HRP_M[FW1-policy-security] quit
```

Step 4 Configure a NAT policy on FW1. After hot standby is enabled, the NAT policy configured on FW1 will be automatically synchronized to FW2.

Configure a NAT policy to translate source addresses on subnet 10.3.0.0/24 to IP addresses in the NAT address pool (1.1.1.2 to 1.1.1.5) when intranet users access the Internet.

```
HRP_M[FW1] nat address-group group1
HRP_M[FW1-address-group-group1] section 0 1.1.1.2 1.1.1.5
HRP_M[FW1-address-group-group1] quit
HRP_M[FW1] nat-policy
HRP_M[FW1-policy-nat] rule name policy_nat1
HRP_M[FW1-policy-nat-rule-policy_nat1] source-zone trust
HRP_M[FW1-policy-nat-rule-policy_nat1] destination-zone untrust
HRP_M[FW1-policy-nat-rule-policy_nat1] source-address 10.3.0.0 24
HRP_M[FW1-policy-nat-rule-policy_nat1] action source-nat address-group group1
```

Step 5 Configure the switches.

Add the three interfaces of SW1 and SW2 to VLAN 10.

Note: use the concept of VLAN. All ports will be access port and default vlan will be vlan10

Verification:

#Run the `display vrrp` command on FW1 to check the status of interfaces in the VRRP group.

```
HRP_M<FW1> display vrrp
GigabitEthernet0/0/1 | Virtual Router 1
  State : Master
  Virtual IP : 10.3.0.3
  Master IP : 10.3.0.1
  PriorityRun : 120
  PriorityConfig : 100
  MasterPriority : 120
```

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```
Preempt : YES    Delay Time : 0 s
TimerRun : 60 s
TimerConfig : 60 s
Auth type : NONE
Virtual MAC : 0000-5e00-0101
Check TTL : YES
Config type : vgmp-vrrp
Backup-forward : disabled
```

```
GigabitEthernet0/0/2 | Virtual Router 2
State : Master
Virtual IP : 1.1.1.1
Master IP : 10.2.0.1
PriorityRun : 120
PriorityConfig : 100
MasterPriority : 120
Preempt : YES    Delay Time : 0 s
TimerRun : 60 s
TimerConfig : 60 s
Auth type : NONE
Virtual MAC : 0000-5e00-0102
Check TTL : YES
Config type : vgmp-vrrp
Backup-forward : disabled
```

Run the **display vrrp** command on FW2 to check the status of interfaces in the VRRP group.

```
HRP_S<FW2>display vrrp
GigabitEthernet0/0/1 | Virtual Router 1
State : Backup
Virtual IP : 10.3.0.3
Master IP : 10.3.0.1
PriorityRun : 120
PriorityConfig : 100
MasterPriority : 120
Preempt : YES    Delay Time : 0 s
TimerRun : 60 s
TimerConfig : 60 s
Auth type : NONE
Virtual MAC : 0000-5e00-0101
Check TTL : YES
Config type : vgmp-vrrp
Backup-forward : disabled

GigabitEthernet0/0/2 | Virtual Router 2
State : Backup
Virtual IP : 1.1.1.1
Master IP : 0.0.0.0
PriorityRun : 120
PriorityConfig : 100
MasterPriority : 0
Preempt : YES    Delay Time : 0 s
TimerRun : 60 s
```

```
TimerConfig : 60 s
Auth type : NONE
Virtual MAC : 0000-5e00-0102
Check TTL : YES
Config type : vgmp-vrrp
Backup-forward : disabled
```


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Run the **display hrp state verbose** command on FW1 to check the current status of the VGMP group.

```
HRP_M<FW1>display hrp state verbose
Role: active, peer: standby
Running priority: 45000, peer: 45000
Backup channel usage: 0.00%
Stable time: 0 days, 0 hours, 46 minutes
Last state change information: 17:18:08 HRP core state changed, old_state = abnormal(active),
new_state = normal, local_priority = 45000, peer_priority = 45000.

Configuration:
hello interval:          1000ms
preempt:                 60s
mirror configuration:    off
mirror session:          off
track trunk member:      on
auto-sync configuration: on
auto-sync connection-status: on
adjust ospf-cost:         on
adjust ospfv3-cost:       on
adjust bgp-cost:          on
nat resource:             off

Detail information:
    GigabitEthernet0/0/1 vrrp vrid 1: active
    GigabitEthernet0/0/2 vrrp vrid 2: active
                                ospf-cost: +0
                                ospfv3-cost: +0
                                bgp-cost: +0
```

Run the **display hrp state verbose** command on FW2 to check the current status of the VGMP group.

```
HRP_S<FW2>display hrp state verbose
Role: standby, peer: active
Running priority: 45000, peer: 45000
Backup channel usage: 0.00%
Stable time: 0 days, 0 hours, 41 minutes
Last state change information: 17:18:08 HRP core state changed, old_state = abnormal(standby),
new_state = normal, local_priority = 45000, peer_priority = 45000.

Configuration:
hello interval:          1000ms
preempt:                 60s
mirror configuration:    off
mirror session:          off
```

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```
track trunk member:      on
auto-sync configuration:  on
auto-sync connection-status: on
adjust ospf-cost:        on
adjust ospfv3-cost:      on
adjust bgp-cost:         on
nat resource:            off
```

Detail information:

```
GigabitEthernet0/0/1 vrrp vrid 1: standby
GigabitEthernet0/0/2 vrrp vrid 2: standby
                        ospf-cost: +65500
                        ospfv3-cost: +65500
                        bgp-cost: +100
```

Ping PC2 in the Untrust Zone from PC1 in the Trust Zone. Run the **display firewall session table** command on FW1 and FW2 to check sessions.

```
HRP_M<FW1> display firewall session table
Current Total Sessions : 1
icmp VPN: public --> public  10.3.0.10:53419[1.1.1.4:2049] --> 1.1.1.10:2048
```

```
HRP_S<FW2> display firewall session table
Current Total Sessions : 1
icmp VPN: public --> public  10.3.0.10:53419[1.1.1.4:2049] --> 1.1.1.10:2048
```

Note: Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-22BSCYS001. Upload the .topo and .cfg files on MS Teams Assignment.

Questions:

1. Describe the three main types of information synchronized by HRP between the active and standby firewalls. Provide an example for each.

HRP synchronizes **session information, NAT information, and configuration information** between the active and standby firewalls. Session information includes active connections such as TCP or UDP sessions; for example, if a user is downloading a file, the session continues after failover. NAT information includes source and destination NAT mappings; for example, internal IP 10.3.0.10 translated to public IP 1.1.1.2 is shared with the standby firewall. Configuration information includes security and NAT policies; for example, a Trust-to-Untrust permit rule configured on the active firewall is automatically copied to the standby firewall.

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2. A network administrator has enabled HRP but forgot to configure the heartbeat interface. What are the potential consequences of this oversight?

If the heartbeat interface is not configured, the firewalls cannot monitor each other's status. As a result, the standby firewall may not detect the failure of the active firewall, and failover may not occur. This can lead to **network downtime, loss of sessions, and service interruption**, defeating the purpose of high availability.

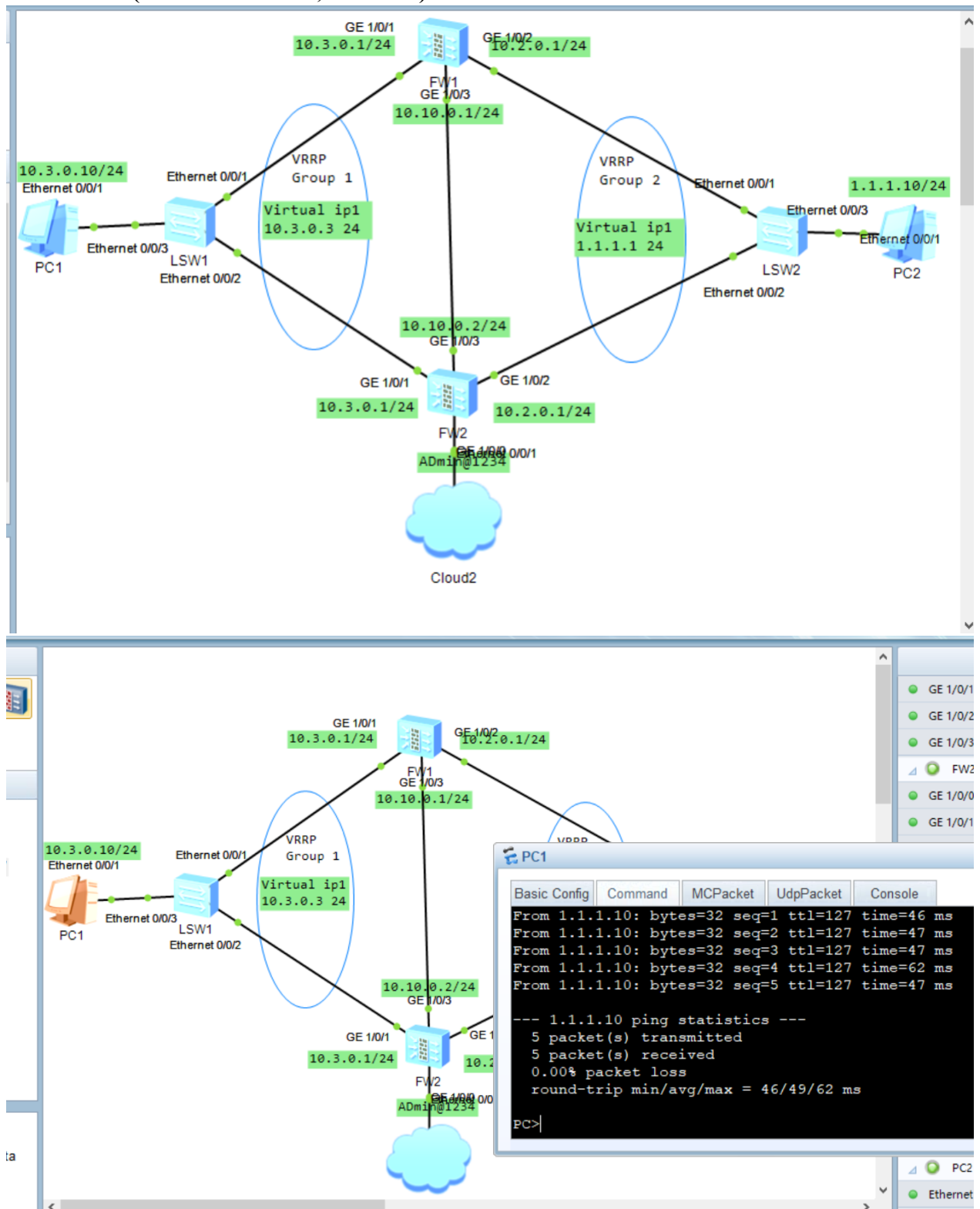
3. Describe the interaction between HRP, VRRP, and VGMP during a typical failover event. Start from the moment a failure is detected on the active firewall and end with the standby firewall taking over.

When the active firewall fails, the heartbeat stops and VGMP detects the failure. VGMP then triggers a role change and forces all VRRP groups on the active firewall to step down. VRRP moves the virtual IP addresses to the standby firewall, making it the new active device. Because HRP has already synchronized session, NAT, and configuration data, the standby firewall takes over smoothly without interrupting ongoing connections.

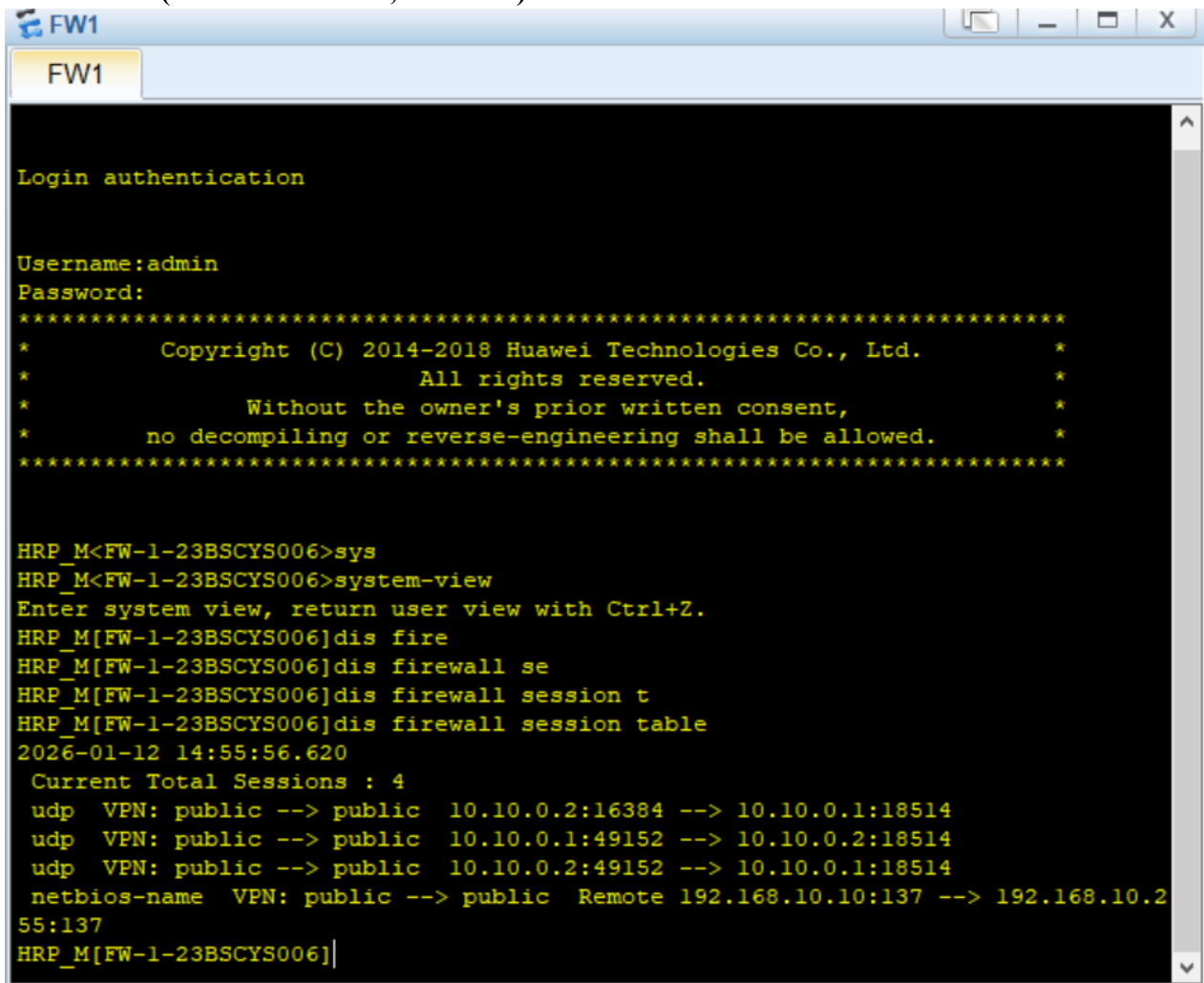
4. Briefly compare and contrast the functions of HRP and VGMP. In a single sentence, summarize the unique contribution of each protocol to the overall high-availability solution.

HRP is responsible for synchronizing sessions, NAT tables, and configurations between firewalls, while VGMP manages and coordinates multiple VRRP groups to ensure a single firewall is active at a time; together, HRP ensures **state continuity** and VGMP ensures **correct role control** in a high-availability setup.

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```
FW1
FW1
Login authentication

Username:admin
Password:
*****
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*                  All rights reserved.                  *
*          Without the owner's prior written consent,          *
*          no decompiling or reverse-engineering shall be allowed.  *
*****

HRP_M<FW-1-23BSCYS006>sys
HRP_M<FW-1-23BSCYS006>system-view
Enter system view, return user view with Ctrl+Z.
HRP_M[FW-1-23BSCYS006]dis fire
HRP_M[FW-1-23BSCYS006]dis firewall se
HRP_M[FW-1-23BSCYS006]dis firewall session t
HRP_M[FW-1-23BSCYS006]dis firewall session table
2026-01-12 14:55:56.620
  Current Total Sessions : 4
    udp  VPN: public --> public  10.10.0.2:16384 --> 10.10.0.1:18514
    udp  VPN: public --> public  10.10.0.1:49152 --> 10.10.0.2:18514
    udp  VPN: public --> public  10.10.0.2:49152 --> 10.10.0.1:18514
    netbios-name  VPN: public --> public  Remote 192.168.10.10:137 --> 192.168.10.2
55:137
HRP_M[FW-1-23BSCYS006]|
```

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